

You cannot make steel without it. Almost no one can name it.

Manganese is the metal hiding inside steel. About 90% of it goes into steelmaking, it has no satisfactory substitute, and at roughly 20 million tonnes a year it is one of the largest metal markets on Earth. Mined in Africa, processed in China, and now wanted for batteries. Here is the data.

90%

Of all manganese goes into making steel.

In the furnace, manganese strips out the **oxygen and sulphur** that make steel brittle, then adds **strength and hardness**. Almost every tonne of steel on Earth contains some, which is why manganese demand tracks the steel industry almost exactly.

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Satisfactory substitutes, in its major use.

This is the line that matters. USGS states plainly that manganese has **no satisfactory substitute** in steelmaking. You can swap one metal for another in many alloys. For this job, there is no replacement, which is rare and makes manganese quietly critical.

One of the biggest metals you never hear about.

20 Mt

of manganese content mined a year, more than zinc, nickel, lead and tin combined.

~10 kg

of manganese in every single tonne of steel produced.

It is produced on the same scale as copper, yet it almost never gets mentioned, because you only ever meet it as part of something else.

An African story.

South Africa about 7.6 million tonnes Mn content	38%
Gabon the strong second	25%
Top 3 (add Ghana) most of world supply	73%

Two African countries supply most of the world's steel ingredient. The risk here is rail lines and ports, not geology.

Mined in Africa, refined in China.

The ore is African, but turning it into usable metal is a different map. China dominates the **ferroalloys** that feed steel mills, and almost entirely controls the **battery-grade manganese sulphate** now wanted for electric-vehicle cathodes.

So the new growth in manganese, batteries, runs straight back through one country again.

Steel made it essential. Batteries make it strategic.

For a century manganese mattered only to steelmakers. Now it is going into lithium-ion cathodes, the nickel-manganese-cobalt (NMC) chemistry and the cheaper lithium-manganese-iron-phosphate (LMFP) cells.

It is a small slice of demand today, but the fastest-growing one, and it puts a steel metal into the heart of the energy transition.

Essential, abundant, and overlooked.

Manganese is the metal nobody worries about until they think it through. It is in nearly every steel, it has no substitute, and the United States mines none of it. The geology is fine. The attention is not.

As steel keeps being made and batteries scale, the quietest metal in the works is one to watch.

Sources: USGS Mineral Commodity Summaries 2026 (manganese). Production on a manganese-content basis; the steel link follows USGS and the World Steel Association.

Next up: **zinc**, the metal that stops the world from rusting.

I'm a materials engineer turned data analyst, breaking down one material at a time: the numbers, the science, and why it matters. Steel, aluminium, copper, rare earths, lithium, nickel, cobalt, titanium, graphite, silicon, silver, tin, magnesium and tungsten came first. Save this, and follow for zinc.

Interactive dashboard + data on GitHub

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